

Systems Biology Modeling

A Brief Tutorial

Instructor:

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Modeling and Simulation of Biochemical Pathways

- Part I - Using MATLAB/Octave for Dynamic Simulation of Biochemical Pathways

References:

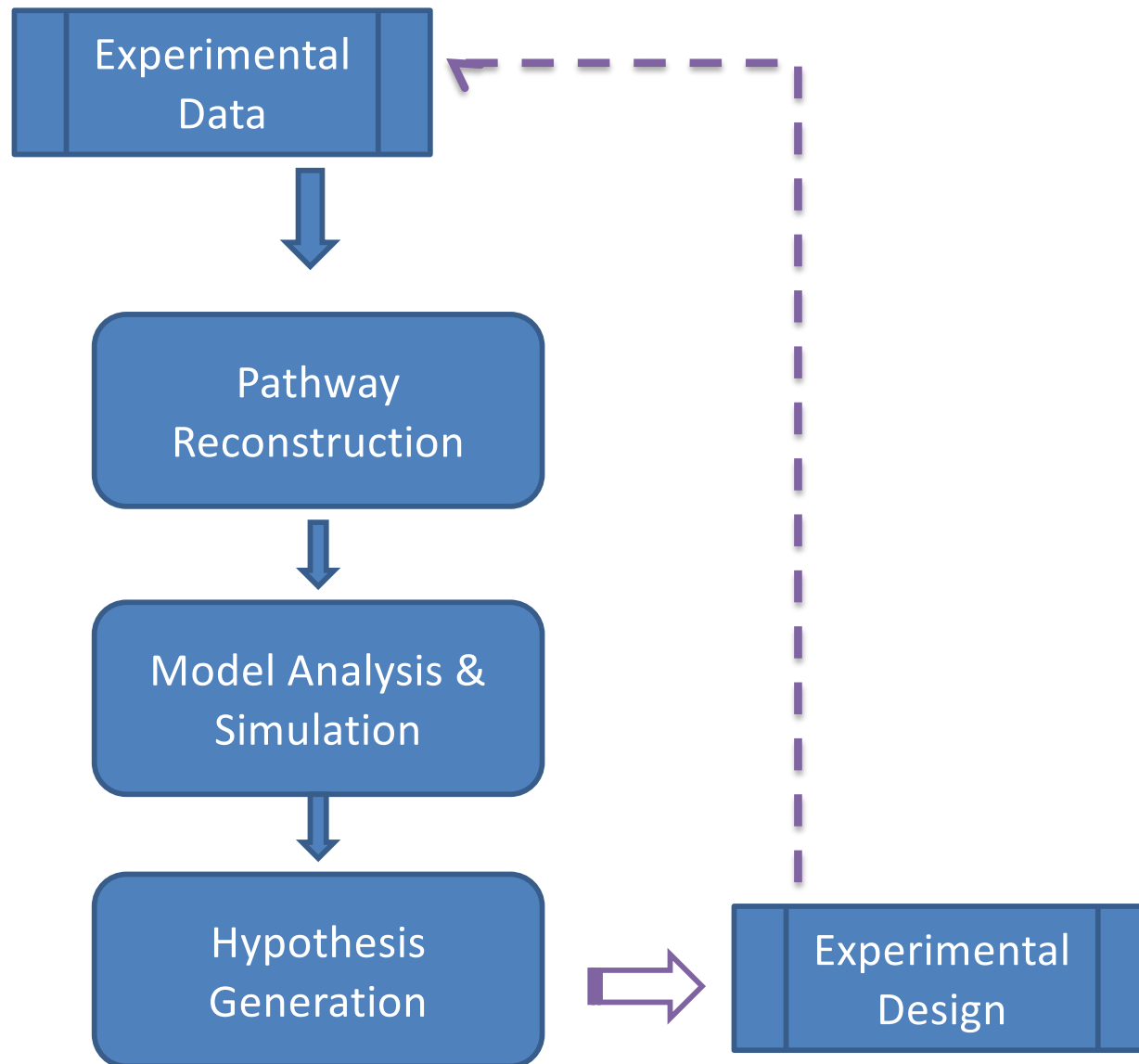
Klipp et al: Sec 3.1, Ch 17

W. J. Palm III, Introduction to Matlab for Engineers, 3rd Ed. (Ch. 9)

Prof. A. Dragomir, BIOE 1397 Lecture Slides

What is Quantitative Systems Biology

What + Where + Who + When = SysBio (HOW) → Why



Systems Biology & Dynamic Modeling

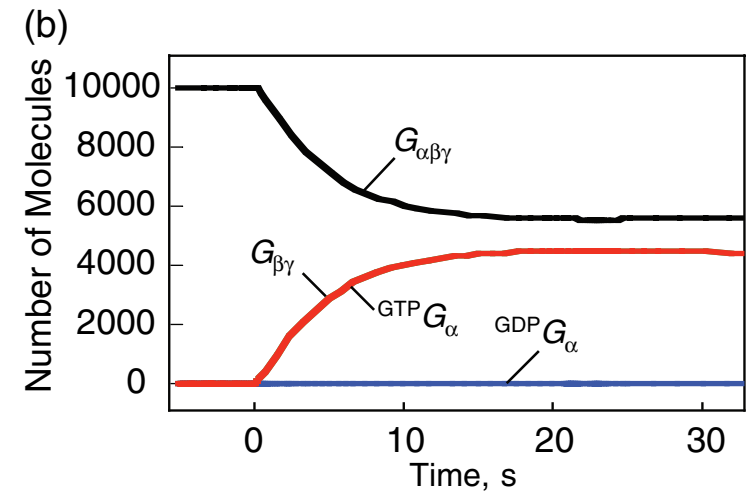
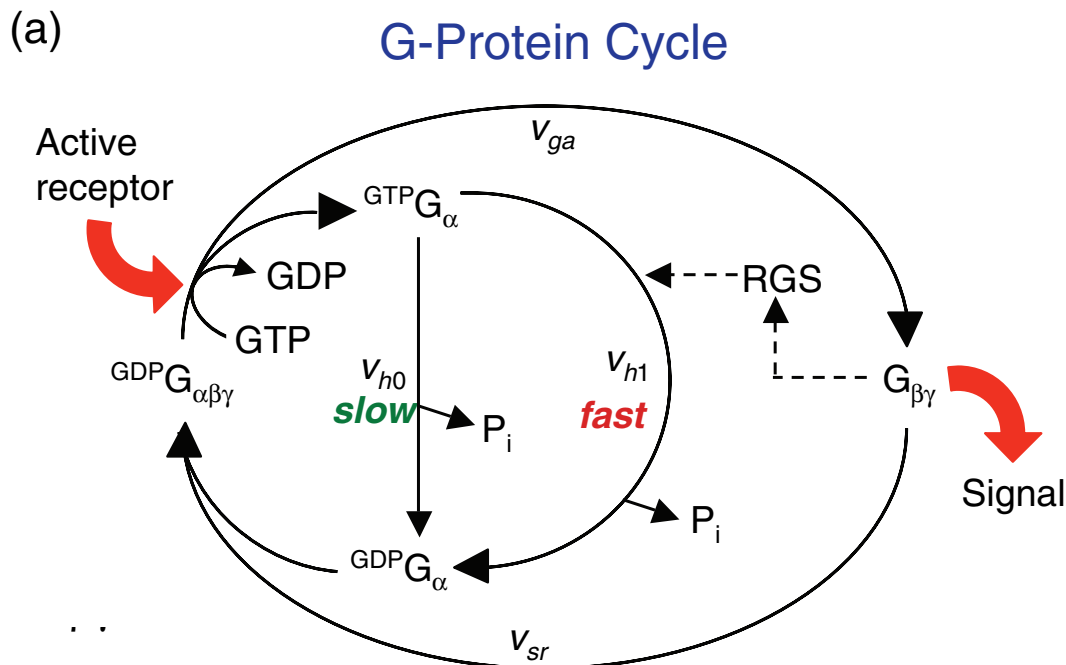
- The time varying behavior of biological systems can be deterministically described by a set of differential equations of the form:
 - $dx_i/dt = f_i(x_1, \dots, x_n, p_1, \dots, p_l, t)$ for $i=1, \dots, n$
 - x_i are the variables (e.g., substrate/product concentrations)
 - p_i are the parameters (e.g., kinetic constants)
 - t is time
- Vector notation
 - $x = [x_1, \dots, x_n]^T$, $f = [f_1, \dots, f_n]^T$, $p = [p_1, \dots, p_l]^T$
- An ordinary differential equation (ODE) describes functions of one independent variable (e.g., time t)
- Partial differential equations (PDEs) describe functions of >1 variables.
- Linear DEs can be written as a linear function of the variables:
 - $dx_i/dt = a_{i1}x_1 + a_{i2}x_2 + \dots$
 - There are no products or powers of the variables $x(t)$ and its derivatives.
 - If the above conditions do not hold, the DE is non-linear.

Systems Biology & Dynamic Modeling

- Ordinary Diff Eqs
 - Can describe biochemical reactions
 - Can describe mRNA abundance using a combination of mass action and Hill equation terms
 - Can use ODE to describe gene transcription and protein production using combination of Hill, M-Menten, or mass action

Modeling Signal Transduction

- G proteins – bind guanine nucleotides GDP and GTP.
 - Fig. 3.13/Ex. 3.2 (Klipp) – example of G protein activation /inactivation



$$\frac{d}{dt} G_{\alpha\beta\gamma} = -v_{ga} + v_{sr}$$

$$\frac{d}{dt} G_{\alpha} GTP = v_{ga} - v_{h0} - v_{h1}$$

$$G_{total\alpha} = G_{\alpha\beta\gamma} + G_{\alpha} GTP + G_{\alpha} GDP$$

$$G_{total\beta\gamma} = G_{\alpha\beta\gamma} + G_{\beta\gamma}$$

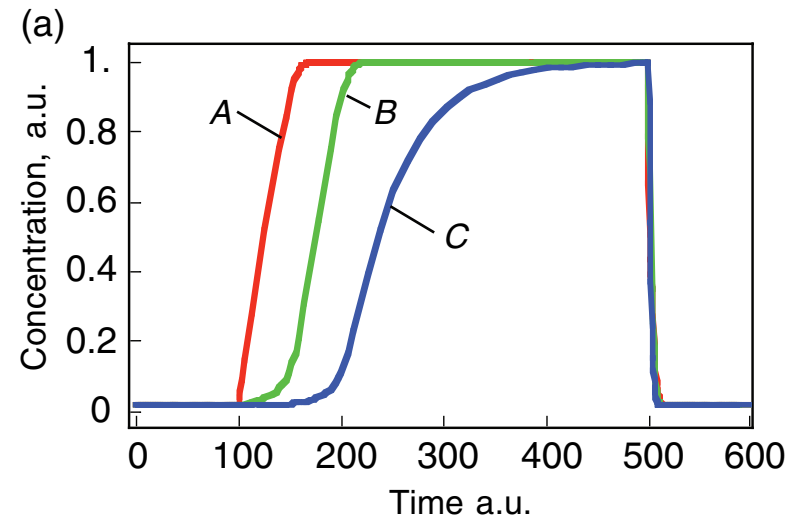
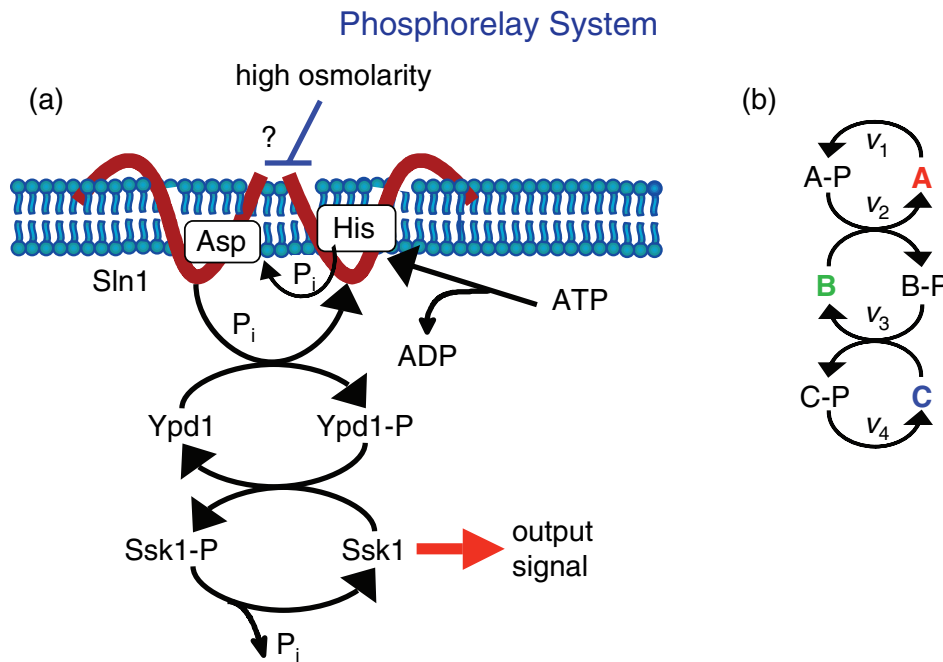
$$v_{ga} = k_{ga} \cdot R_a \cdot G_{\alpha\beta\gamma}$$

$$v_{hi} = k_{hi} \cdot G_{\alpha} GTP, \quad i = 0, 1$$

$$v_{sr} = k_{sr} \cdot G_{\beta\gamma} \cdot G_{\alpha} GDP.$$

Modeling Signal Transduction

- Phosphorelay/Phosphotransfer systems
 - Fig. 3.15 (Klipp, see Sec. 3.2.4.3)



$$\frac{d}{dt}A = -k_1 \cdot A + k_2 \cdot AP \cdot B$$

$$\frac{d}{dt}B = -k_2 \cdot AP \cdot B + k_3 \cdot BP \cdot C$$

$$\frac{d}{dt}C = -k_3 \cdot BP \cdot C + k_4 \cdot CP.$$

$$A_{\text{total}} = A + AP$$

$$B_{\text{total}} = B + BP$$

$$C_{\text{total}} = C + CP.$$

Modeling Signal Transduction

- MAP Kinase Cascades
 - Mitogen-activated protein kinases
 - Family of serine/threonine kinases that transduce signals from cell membrane to nucleus
 - Consists of several levels where the activated kinase at each level phosphorylates the kinase at the next level down the cascade
 - Dynamics represented by ODE derived using mass action kinetics
 - Include conservation equations for MAPK proteins (Klipp, Sec. 3.2.4.4)
 - If phosphate donor groups (ATP/ADP) considered explicitly, include conservation equations for these as well
- Jak/Stat Pathway

Simulating Dynamic Systems with MATLAB/Octave

(b) Design of Structured Dynamic Models

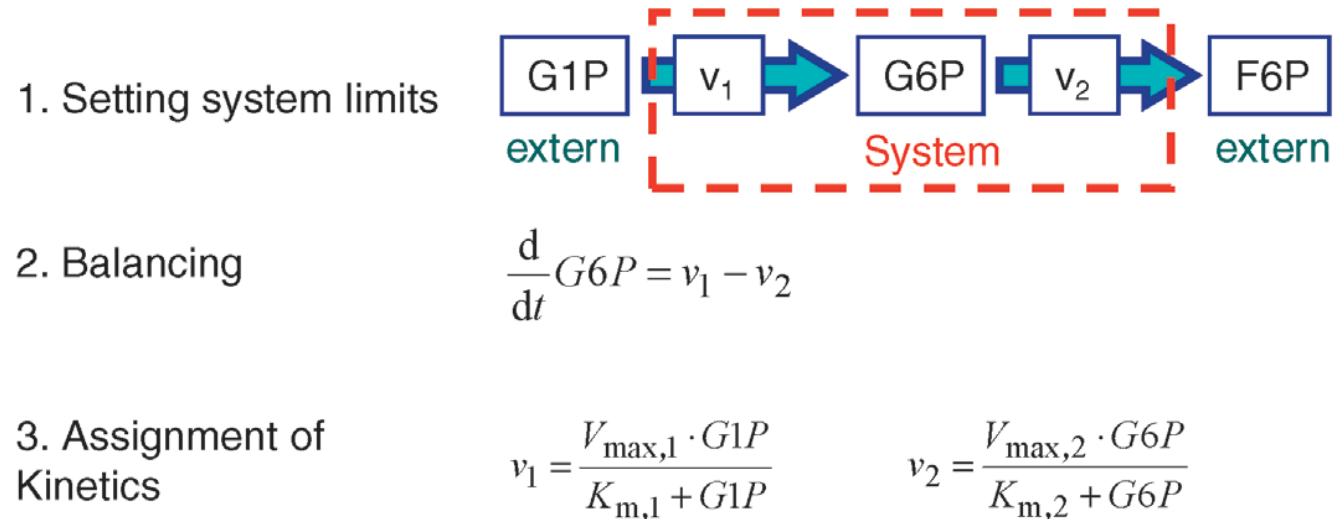
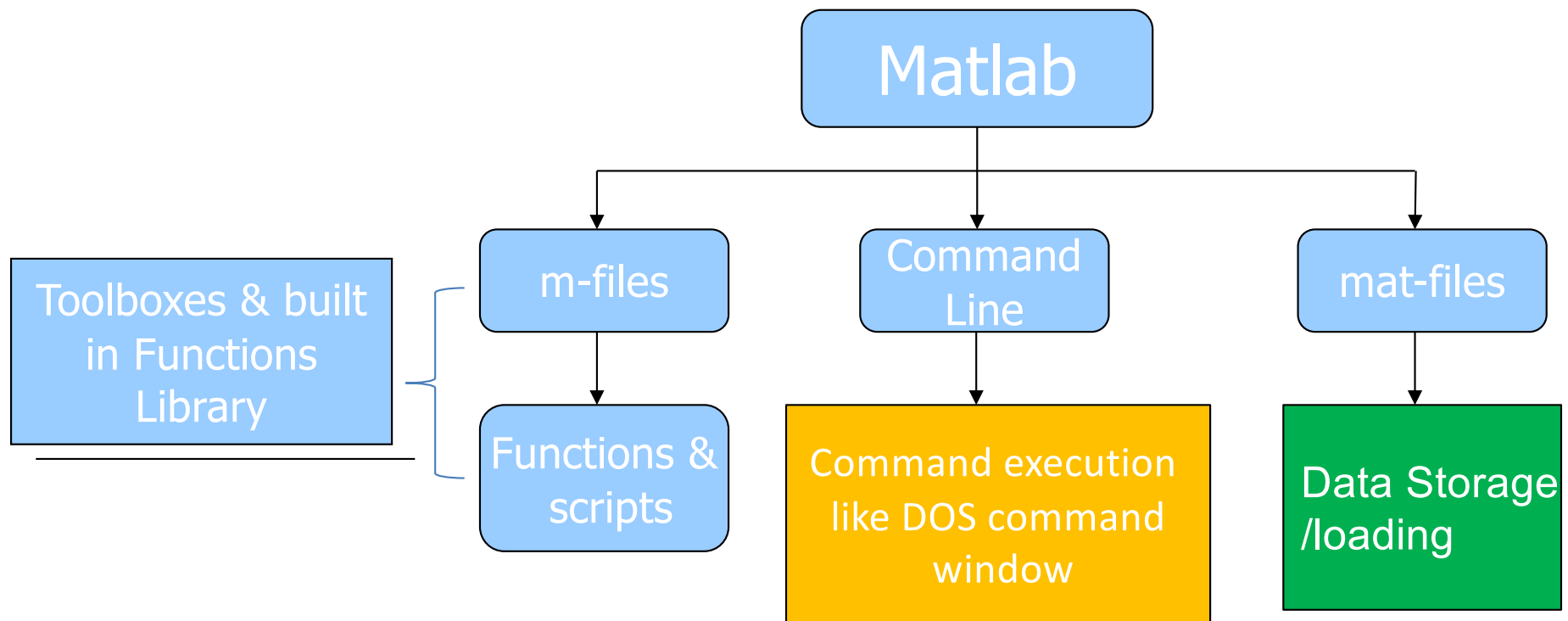


Figure 3.1 Designing metabolic models. (a) Basic elements of metabolic networks and (b) basic steps for designing structured dynamic models (see the text for further explanation).

Introduction to MATLAB/OCTAVE

MATLAB Overview

- Extensible using "Toolkits" (Toolboxes) or user-contributed programs called M-files.
- Built-in functionality includes: matrix manipulation and linear algebra, Data analysis, graphics and visualization

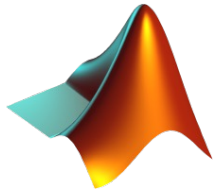


The MATLAB System

MATLAB system consists of main parts:

- **Desktop Tools and Development Environment**
 - Includes the MATLAB desktop and Command Window, an editor and debugger, a code analyzer, browsers for viewing help, the workspace, files, and other tools
- **Mathematical Function Library**
 - vast collection of computational algorithms ranging from elementary functions, like sine, cosine, and complex arithmetic, to more sophisticated functions like matrix inverse, matrix eigenvalues, Bessel functions, and fast Fourier transforms etc.

MATLAB Desktop



The MATLAB Desktop interface is shown with the following components:

- Current Directory:** c:\MyMATLABFiles
- Command Window:** Contains the following commands and outputs:

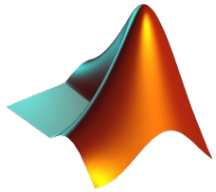
```
>> clear
>> A=2*5^3
A =
    250
>> B=exp(0.005*A)
B =
    3.4903
>> C=4*sqrt(A+B^3)
C =
    68.4130
>> D=5*log10(1000)
D =
    15
>> x = 0:0.02:9;
>> y = A*sin(3*x);
>> plot(x,y)
>> |
```
- Workspace:** Displays the following variables:

Name	Value	Min	Max
A	250	250	250
B	3.4903	3.4903	3.4903
C	68.4130	68.4130	68.4130
D	15	15	15
x	<1x451 double>	0	9
y	<1x451 double>	-249....	249.9...
- Command History:** Shows the following commands:

```
clc
clear
A=2*5^3
B=exp(0.05*A)
C=4*sqrt(A^3+B^3)
D=5*log10(1000)
x = 0:0.03:9
clc1
clc
clear
A=2*5^3
B=exp(0.005*A)
C=4*sqrt(A+B^3)
D=5*log10(1000)
x = 0:0.02:9;
y = A*sin(3*x);
plot(x,y)
```

Command window:
□ type commands

MATLAB Desktop



The MATLAB Desktop interface is shown with the following panes:

- Current Directory:** Displays a list of files in the current directory, including `actuator.m`, `cost.m`, `CubicEval.m`, `cubicsprin...`, `dcmotor.m`, `ethanol.m`, `Falling_Sp...`, `friction1.mdl`, `half-sine.m`, `height.m`, `perfspecs.m`, `plantcost.m`, `speed.m`, `TorqueDat...`, `trapezoid.m`, and `trebuchet...`.
- Command Window:** Shows the MATLAB command prompt with the following commands and outputs:

```
>> clear
>> A=2*5^3
A =
    250
>> B=exp(0.005*A)
B =
    3.4903
>> C=4*sqrt(A+B^3)
C =
    68.4130
>> D=5*log10(1000)
D =
    15
>> x = 0:0.02:9;
>> y = A*sin(3*x);
>> plot(x,y)
>>
```
- Workspace:** Displays a table of variables in the workspace. A red box highlights this pane, and a red arrow points from the `plot(x,y)` command in the Command Window to this pane.
- Command History:** Shows a list of commands entered in the Command Window, including `clc`, `clear`, `A=2*5^3`, `B=exp(0.05*A)`, `C=4*sqrt(A^3+B^3)`, `D=5*log10(1000)`, `x = 0:0.03:9`, `clc`, `clear`, `A=2*5^3`, `B=exp(0.005*A)`, `C=4*sqrt(A+B^3)`, `D=5*log10(1000)`, `x = 0:0.02:9;`, `y = A*sin(3*x);`, and `plot(x,y)`.

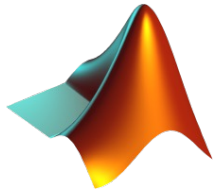
Workspace:

Name	Value	Min	Max
A	250	250	250
B	3.4903	3.4903	3.4903
C	68.4130	68.4130	68.4130
D	15	15	15
x	<1x451 double>	0	9
y	<1x451 double>	-249....	249.9...

Workspace:

- View program variables
- Double click on a variable to see it in the Array Editor

MATLAB Desktop



File Edit Debug Desktop Window Help

Current Directory: c:\MyMATLABFiles

Shortcuts How to Add What's New

Current Directory

Name	Date Modified
actuator.m	8/7/09 1:56 PM
cost.m	8/7/09 1:56 PM
CubicEval.m	8/7/09 1:07 PM
cubicsprin...	8/7/09 1:57 PM
dcmotor.m	8/7/09 1:57 PM
ethanol.m	8/7/09 1:57 PM
Falling_Sp...	3/1/04 4:49 AM
friction1.mdl	8/7/09 1:57 PM
half-sine.m	6/26/05 4:45 PM
height.m	8/7/09 1:58 PM
perfspecs.m	10/16/03 5:46 PM
plantcost.m	6/29/04 7:11 AM
speed.m	7/13/03 6:33 AM
TorqueDat...	10/18/04 8:07 AM
trapezoid.m	10/3/03 5:15 AM
trebuchet...	7/10/03 5:34 AM

Command Window

```
>> clear
>> A=2*5^3
A =
    250
>> B=exp(0.005*A)
B =
    3.4903
>> C=4*sqrt(A+B^3)
C =
    68.4130
>> D=5*log10(1000)
D =
    15
>> x = 0:0.02:9;
>> y = A*sin(3*x);
>> plot(x,y)
>>
```

Workspace

Name	Value	Min	Max
A	250	250	250
B	3.4903	3.4903	3.4903
C	68.4130	68.4130	68.4130
D	15	15	15
x	<1x451 double>	0	9
y	<1x451 double>	-249....	249.9...

Command History

```
8/7/09 12:49 PM
clear
clear
A=2*5^3
B=exp(0.005*A)
C=4*sqrt(A+B^3)
D=5*log10(1000)
x = 0:0.02:9
y = A*sin(3*x);
plot(x,y)
clear
clear
A=2*5^3
B=exp(0.005*A)
C=4*sqrt(A+B^3)
D=5*log10(1000)
x = 0:0.02:9;
y = A*sin(3*x);
plot(x,y)
```

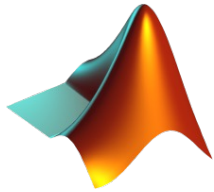
Start

OVR

Current Directory

View folders

MATLAB Desktop



The screenshot shows the MATLAB Desktop environment. The Command Window is active, displaying the following commands and their outputs:

```
>> clear
>> A=2*5^3
A =
    250
>> B=exp(0.005*A)
B =
    3.4903
>> C=4*sqrt(A+B^3)
C =
    68.4130
>> D=5*log10(1000)
D =
    15
>> x = 0:0.02:9;
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>> plot(x,y)
>> |
```

The Workspace window shows the following variables:

Name	Value	Min	Max
A	250	250	250
B	3.4903	3.4903	3.4903
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D	15	15	15
x	<1x451 double>	0	9
y	<1x451 double>	-249....	249.9...

The Command History window is highlighted with a red border and a red arrow pointing to it from the Command Window. It shows the following commands:

```
%-- 8/7/09 12:49 PM --%
clear
clear
A=2*5^3
B=exp(0.05*A)
C=4*sqrt(A^3+B^3)
D=5*log10(1000)
x = 0:0.03:9
clc
clear
A=2*5^3
B=exp(0.005*A)
C=4*sqrt(A+B^3)
D=5*log10(1000)
x = 0:0.02:9;
y = A*sin(3*x);
plot(x,y)
```

The Current Directory window shows the following files:

Name	Date Modified
actuator.m	8/7/09 1:56 PM
cost.m	8/7/09 1:56 PM
CubicEval.m	8/7/09 1:07 PM
cubicsprin...	8/7/09 1:57 PM
dcmotor.m	8/7/09 1:57 PM
ethanol.m	8/7/09 1:57 PM
Falling_Sp...	3/1/04 4:49 AM
friction1.mdl	8/7/09 1:57 PM
half-sine.m	6/26/05 4:45 PM
height.m	8/7/09 1:58 PM
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speed.m	7/13/03 6:33 AM
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trapezoid.m	10/3/03 5:15 AM
trebuchet...	7/10/03 5:34 AM

The Command History window is highlighted with a red border and a red arrow pointing to it from the Command Window. It shows the following commands:

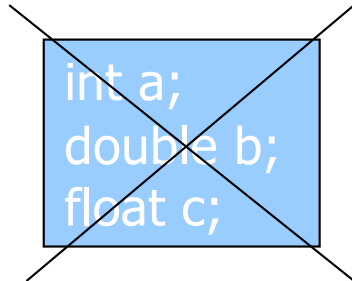
```
%-- 8/7/09 12:49 PM --%
clear
clear
A=2*5^3
B=exp(0.05*A)
C=4*sqrt(A^3+B^3)
D=5*log10(1000)
x = 0:0.03:9
clc
clear
A=2*5^3
B=exp(0.005*A)
C=4*sqrt(A+B^3)
D=5*log10(1000)
x = 0:0.02:9;
y = A*sin(3*x);
plot(x,y)
```

Command History

- ❑ ↑, ↓ keys in the command window view past commands

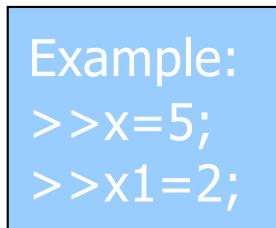
Variables

- No need for types. i.e.,



```
int a;  
double b;  
float c;
```

- All variables are created with double precision unless specified and they are matrices.



```
Example:  
>>x=5;  
>>x1=2;
```

- After these statements, the variables are scalars (1x1 matrices) with double precision

Scripts and Functions

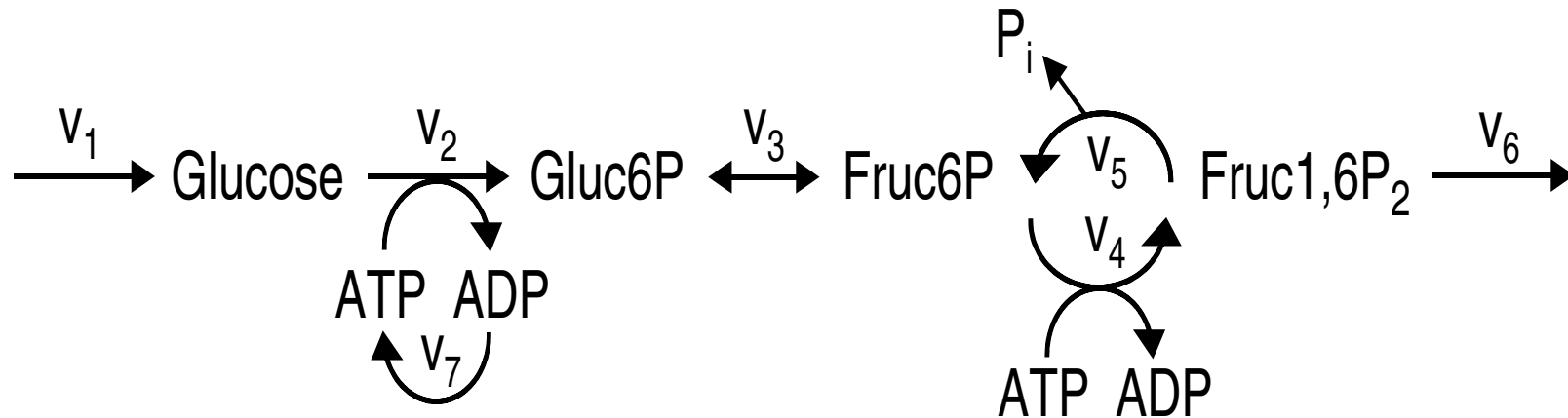
- There are two kinds of M-files:
 - **Scripts**, which do not accept input arguments or return output arguments. They operate on data in the workspace. Any variables that they create remain in the workspace, to be used in subsequent computations
 - **Functions**, which can accept input arguments and return output arguments. Internal variables are local to the function.

Simulating Dynamic Systems with MATLAB/Octave

- Example: Simplified model of upper glycolysis (Klipp et al., Fig. 3.2, Eq. 3.1)
 1. Write the system ODEs for each substrate in term of system fluxes (v_1 , v_2 , etc)
 2. Describe the kinetics, rate equation for each system flux
 3. Determine all the parameters and initial conditions for the system
 4. Translate your system into a Matlab function described in an M file.
 5. Use Matlab ODE solver to run your model and plot results.
 - Commands can be run from the command line or using a script.

Simulating Dynamic Systems

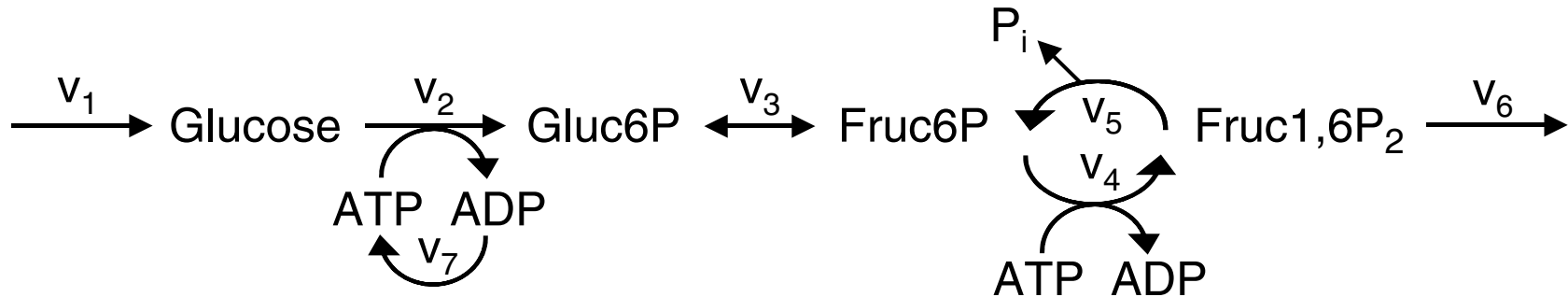
- Example: Simplified model of upper glycolysis (Klipp et al., Fig. 3.2, Eq. 3.1) **Steps 1 – 3:**



- Variables?
- $\text{Glu} = +v_1 - v_2$
- $\text{Glu6} = +v_2 - v_3$
- ...

Simulating Dynamic Systems

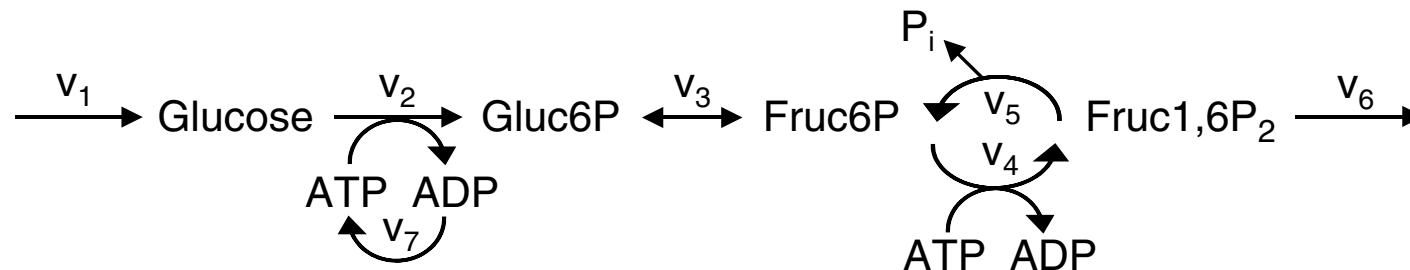
- Example: Simplified model of upper glycolysis (Klipp et al., Fig. 3.2, Eq. 3.1) **Steps 1 – 3:**



With mass action kinetics, the rate equations read $v_1 = \text{const.} = k_1$, $v_2 = k_2 \cdot \text{Glucose} \cdot \text{ATP}$, $v_3 = k_3 \cdot \text{Gluc6P} - k_{-3} \cdot \text{Fruc6P}$, $v_4 = k_4 \cdot \text{Fruc6P} \cdot \text{ATP}$, $v_5 = k_5 \cdot \text{Fruc1,6P}_2$, $v_6 = k_6 \cdot \text{Fruc1,6P}_2$, and $v_7 = k_7 \cdot \text{ADP}$. Given the values of the parameters k_i , $i = 1, \dots, 7$

Simulating Dynamic Systems

- Example: Simplified model of upper glycolysis (Klipp et al., Fig. 3.2, Eq. 3.1) **Steps 1 – 3:**



$$\frac{d}{dt} \text{Glucose} = v_1 - v_2$$

$$\frac{d}{dt} \text{Gluc6P} = v_2 - v_3$$

$$\frac{d}{dt} \text{Fruc6P} = v_3 - v_4 + v_5$$

$$\frac{d}{dt} \text{Fruc1,6P}_2 = v_4 - v_5 - v_6$$

$$\frac{d}{dt} \text{ATP} = -\frac{d}{dt} \text{ADP} = -v_2 - v_4 + v_7.$$

With mass action kinetics, the rate equations read $v_1 = \text{const.} = k_1$, $v_2 = k_2 \cdot \text{Glucose} \cdot \text{ATP}$, $v_3 = k_3 \cdot \text{Gluc6P} - k_{-3} \cdot \text{Fruc6P}$, $v_4 = k_4 \cdot \text{Fruc6P} \cdot \text{ATP}$, $v_5 = k_5 \cdot \text{Fruc1,6P}_2$, $v_6 = k_6 \cdot \text{Fruc1,6P}_2$, and $v_7 = k_7 \cdot \text{ADP}$. Given the values of the parameters k_i , $i = 1, \dots, 7$

Simulating Dynamic Systems with MATLAB/Octave

- Example: Simplified model of upper glycolysis (Klipp et al., Fig. 3.2, Eq. 3.1)
 1. Write the system ODEs for each substrate in term of system fluxes ($v1$, $v2$, etc)
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 3. Determine all the parameters and initial conditions for the system
 4. Translate your system into a Matlab function described in an M file.
 5. Use Matlab ODE solver to run your model and plot results.
 - Commands can be run from the command line or using a script.

Simulating Dynamic Systems with MATLAB/Octave

- Matlab has several solvers that implement various algorithms (e.g. Runge-Kutta, or Euler) to generate numerical solutions for ODEs
- Algorithms are adaptive and use variable step sizes (larger steps when solution is changing more slowly and smaller step sizes in more variable regions)
- Several solvers, two recommended:
 - ode45 – general purpose solver
 - Ode15s – good for difficult or stiff equations
 - If ode45 results in lengthy simulation or error, try ode15s

Simulating Dynamic Systems with MATLAB/Octave

- Basic syntax:
 - `[t,y] = ode45(@ydot, tspan, y0);`
 - `@ydot` = the handle to the ODE function (name of the M-file where ODE system defined)
 - `tspan = [t0, tfinal]` simulation time span
 - `y0 = [yInitial]` vector of initial values for your system
 - `t` = time points for values returned by the solver
 - `y` = corresponding solution(s)/values for the system

Simulating Dynamic Systems with MATLAB/Octave

- Can set options for the solver
 - Options=odeset('name_option1','value1','name_option2','value2',....)
 - Name=some property to be set; value=assigned value
 - In Matlab command window, type “odeset” to see possible values to set or “help odeset” for descriptions of the options
 - ‘Refine’ – produces more points and results in smoother solution curves/plots.
 - ‘AbsTol’, ‘RelTol’ – can set integration error tolerance based on the range of values your system will return.

Simulating Dynamic Systems with MATLAB

- Example: Simplified model of upper glycolysis (Klipp et al., Fig. 3.2, Eq. 3.1)
- M file **(Step 4)**

```
function dydt=glu3(t,y,k1,k2,k3,nk3,k4,k5,k6,k7)
%function dydt=glu3(t,y,k1,k2,k3,nk3,k4,k5,k6,k7)
% glu3 Ex chpt 3
% Created by E May on 4/2013.
```

```
%for indexing
glu=1; g6p=2; f6p=3;f16p2=4; atp=5;adp=6;
%rate eqs
v1=k1;
v2=k2*y(glu)*y(atp);
v3=k3*y(g6p) - nk3*y(f6p);
v4=k4*y(f6p)*y(atp);
v5=k5*y(f16p2);
v6=k6*y(f16p2);
v7=k7*y(adp);
```

```
%ODE for network
y(glu)=v1-v2;
y(g6p)=v2-v3;
y(f6p)=v3-v4+v5;
y(f16p2)=v4-v5-v6;
y(atp)=-v2-v4+v7;
y(adp)=v2+v4-v7;

dydt=y;
```

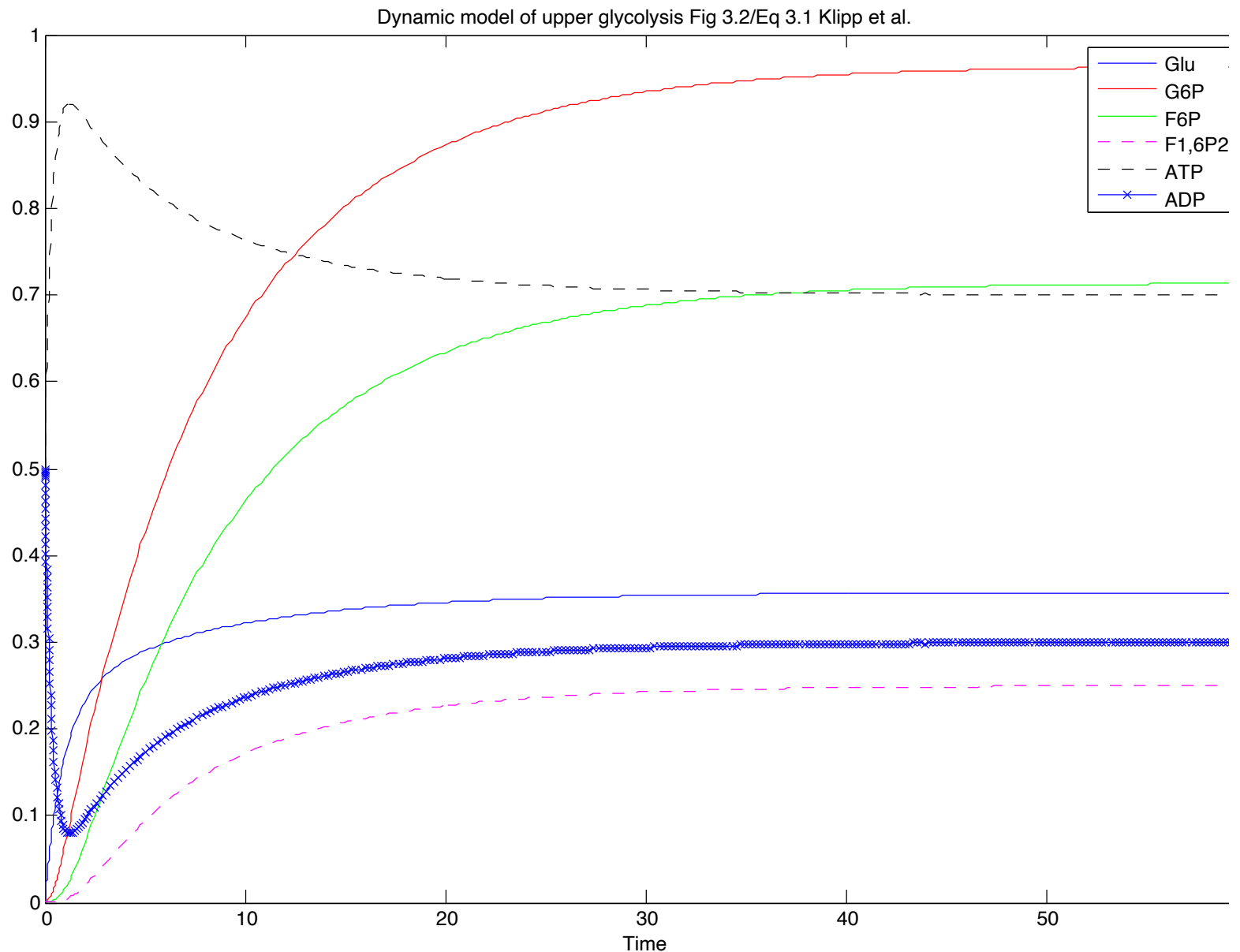
Simulating Dynamic Systems with MATLAB

- **Step 5:**
- Can capture command line instructions using 'diary' command:
- `>>diary 'filename.log'`

```
options=odeset('OutputFcn',[],'AbsTol',1e-10, 'RelTol', 1e-3);
[t,y]=ode45(@glu3,[0,60],[0,0,0,0,0.5,0.5],options,0.25,1,1,1,1,1,1,2.5);
plot(t,y(:,1))
hold
Current plot held
plot(t,y(:,2),'r')
plot(t,y(:,3),'g')
plot(t,y(:,4),'m--')
plot(t,y(:,5),'k--')
plot(t,y(:,6),'x-')
title('Dynamic model of upper glycolysis Fig 3.2/Eq 3.1 Klipp et al.')
ylabel('Concentration')
xlabel('Time')
legend('Glu','G6P','F6P','F1,6P2','ATP','ADP')
diary off
```

Simulating Dynamic Systems with MATLAB

- Plot



HANDS ON PRACTICE: Model Simulation with OCTAVE*

*Thanks to Daniel Ajuzie for Octave scripts

Model Simulation with OCTAVE

Model/Function File: glu3.m

```
function dydt = glu3(t, y, k1, k2, k3, nk3, k4, k5,  
k6, k7)
```

```
% For indexing
```

```
glu = 1;
```

```
g6p = 2;
```

```
f6p = 3;
```

```
f16p2 = 4;
```

```
atp = 5;
```

```
adp = 6;
```

```
% Rate equations
```

```
v1 = k1;
```

```
v2 = k2 * y(glu) * y(atp);
```

```
v3 = k3 * y(g6p) - nk3 * y(f6p);
```

```
v4 = k4 * y(f6p) * y(atp);
```

```
v5 = k5 * y(f16p2);
```

```
v6 = k6 * y(f16p2);
```

```
v7 = k7 * y(adp);
```

```
% ODE equations
```

```
dydt(glu) = v1 - v2;
```

```
dydt(g6p) = v2 - v3;
```

```
dydt(f6p) = v3 - v4 + v5;
```

```
dydt(f16p2) = v4 - v5 - v6;
```

```
dydt(atp) = -v2 - v4 + v7;
```

```
dydt(adp) = v2 + v4 - v7;
```

```
dydt = dydt';
```

```
End
```

Model Simulation with OCTAVE

Execution/Script File: solve_glu.m

```
options = odeset('OutputFcn', [], 'AbsTol', 1e-10, 'RelTol', 1e-3);  
[t, y] = ode45(@(t, y) glu3(t, y, 0.25, 1, 1, 1, 1, 1, 1, 2.5), [0, 60],  
[0, 0, 0, 0, 0.5, 0.5], options);  
  
plot(t, y(:, 1));  
hold on;  
plot(t, y(:, 2), 'r');  
plot(t, y(:, 3), 'g');  
plot(t, y(:, 4), 'm');  
plot(t, y(:, 5), 'k');  
plot(t, y(:, 6), 'x-');  
title('Upper glycolysis model');  
ylabel('concentration');  
xlabel('time');  
legend('Glu', 'G6P', 'F6P', 'F16P2', 'ATP', 'ADP');
```

Simulating Dynamic Systems with Matlab

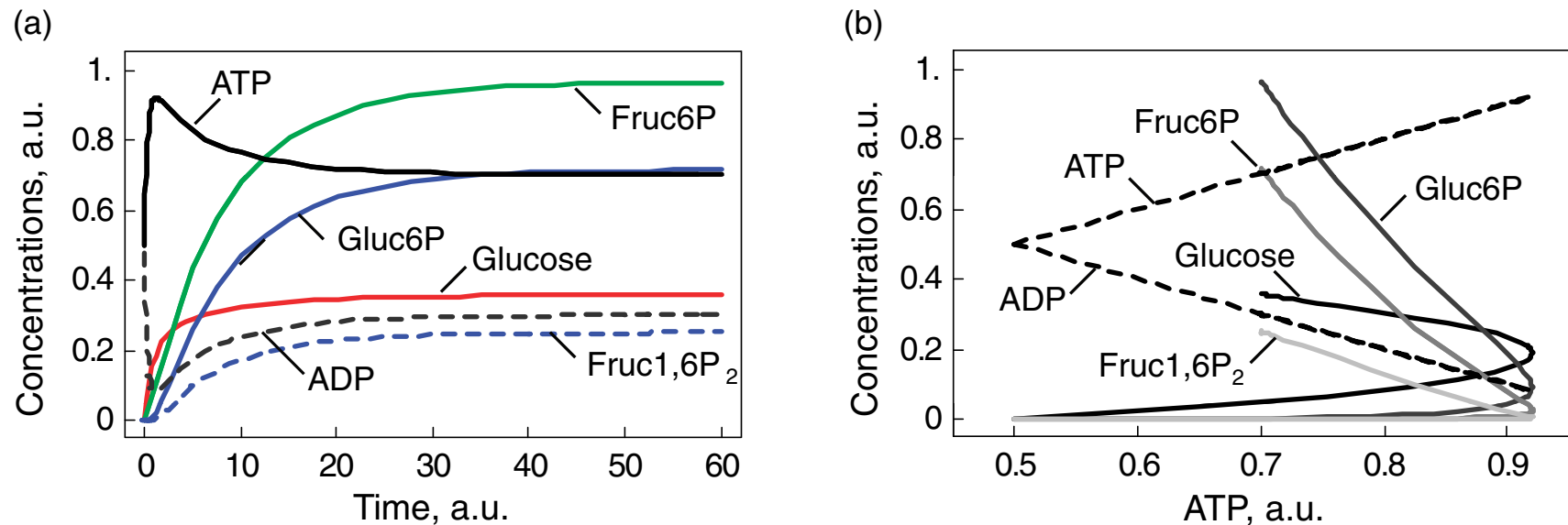


Figure 3.3 Dynamic behavior of the upper glycolysis model (Figure 3.2 and Eq. (3.1)). Initial conditions at $t = 0$: $Glucose(0) = Gluc-6P(0) = Fruc-6P(0) = Fruc-1,6P_2(0) = 0$ and $ATP(0) = ADP(0) = 0.5$ (arbitrary units). Parameters: $k_1 = 0.25$, $k_2 = 1$, $k_3 = 1$, $k_4 = 1$, $k_5 = 1$, $k_6 = 1$, and $k_7 = 2.5$. The steady-state concentrations are $Glucose^{st} = 0.357$, $Gluc-6P^{st} = 0.714$, $Fruc-6P^{st} =$

0.964 , $Fruc-1,6P_2^{st} = 0.2$, and $ATP^{st} = 0.7$, and $ADP^{st} = 0.25$. The steady-state fluxes are $J_1 = J_2 = J_3 = J_5 = J_6 = 0.25$, $J_4 = 0.5$, and $J_7 = 0.75$. (a) Time-course plots (concentration versus time), (b) phase-plane plot (concentrations versus concentration of ATP with varying time); all curves start at $ATP = 0.5$ for $t = 0$.

Systems Biology Modeling Tutorial

Instructor:

Dr. Elebeoba E. May, PhD (emay5@wisc.edu)

Medical Microbiology & Immunology Department

Wisconsin Institute of Discovery

University of Wisconsin-Madison

Modeling and Simulation of Biochemical Pathways

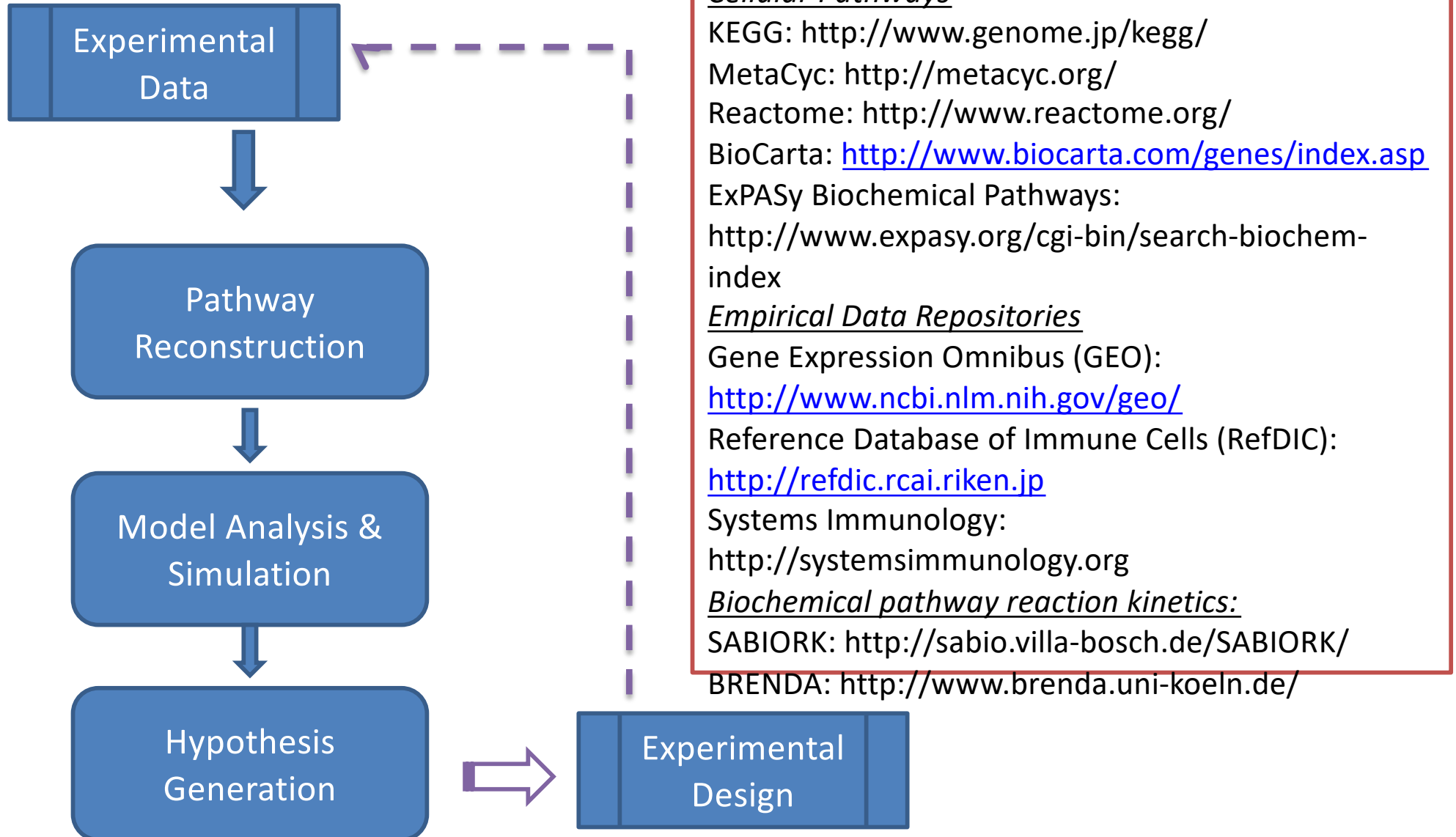
- Part II – Resources for Systems Biology Model Development

Data Resources for Model Development

- Model development requires various types of data
 - Model components
 - Functions and interaction of components
 - Reaction kinetics
 - Primary data resources
 - Organism specific data resources

What is Quantitative Systems Biology

What + Where + Who + When = SysBio (HOW) → Why



KEGG DB

- www.kegg.com (*Kyoto Encyclopedia of Genes & Genomes*)
- Contains information about genes, proteins, biochemical compounds, reactions, and pathways,
- Organized in three parts:
 - Gene universe: GENES, SSDB (seq sim), KO (ortholog DB)
 - Chemical universe

CHEMICAL Universe (text/table from KEGG)

KEGG LIGAND contains our knowledge on the universe of chemical substances and reactions that are relevant to life. It is a composite database consisting of COMPOUND, GLYCAN, REACTION, RPAIR, RCLASS, and ENZYME databases, [...](#)

KEGG COMPOUND is a collection of small molecules, biopolymers, and other chemical substances that are relevant to biological systems.

KEGG REACTION contains all reactions taken from **KEGG ENZYME** and additional reactions taken from the metabolic pathway maps in KEGG PATHWAY.

Database		Identifier	Content	Specialized entry point
LIGAND	COMPOUND	C number	Chemical compound structures	KEGG COMPOUND
	GLYCAN	G number	Glycan structures	KEGG GLYCAN
	REACTION	R number	Biochemical reactions	KEGG REACTION
	RPAIR	RP number	Reactant pair alignments	
	RCLASS	RC number	Reaction class	
	ENZYME	EC number	Enzyme nomenclature	

KEGG DB

- KEGG DB hierarchical classification
 - Metabolism
 - Genetic information processing
 - Environmental information processing
 - Cellular processes
 - Human diseases
- Maps are manually drawn
- KEGG can be queried via web or data downloaded via FTP site



KEGG - Table of Contents

Data Types and Resources
www.kegg.com (Kyoto Encyclopedia of Genes & Genomes)

KEGG2 PATHWAY BRITE MODULE DISEASE DRUG KO GENOME GENES LIGAND DBGET

Search for

Category	Entry Point	Release Info	Search & Compute	DBGET Search
Systems information	KEGG PATHWAY	New maps	Search Pathway	PATHWAY BRITE MODULE
	KEGG BRITE	Update history	Search Brite	
	KEGG MODULE	New hierarchies	Search Module	
	KEGG Mapper	Update history	KEGG pathway maps	
	KEGG Atlas		BRITE functional hierarchies	
			KEGG modules	
	KEGG DISEASE	New drug maps Update history	Human diseases	DISEASE DRUG ENVIRON
	KEGG DRUG		Infectious diseases	
	KEGG ENVIRON		ATC drug classification	
	KEGG MEDICUS			
Genomic information	KEGG ORTHOLOGY		KEGG Orthology (KO)	ORTHOLOGY
	KEGG GENES	New organisms Update history	SSDB search	GENES
	KEGG GENOME		BLAST / FASTA search	DGENES
	KEGG Organisms		KAAS automatic annotation	EGENES
			Map organisms to taxonomy	MGENES
			Generate taxonomy tree	GENOME
Chemical information	KEGG LIGAND		KEGG organisms	EGENOME
	KEGG COMPOUND			MGENOME
	KEGG GLYCAN		SIMCOMP / SUBCOMP search	COMPOUND
	KEGG REACTION		KCaM search	GLYCAN
	Reaction Modules		E-zyme reaction prediction	REACTION
			PathPred pathway prediction	RPAIR
			PathComp path computation	RCLASS
			PathSearch reaction search	ENZYME

See Kanehisa et al. (2012) for the new features of KEGG.

Data Types and Resources

www.kegg.com (*Kyoto Encyclopedia of Genes & Genomes*)

DBGET Search Result: ENZYME isocitrate lyase - Mozilla Firefox

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http://www.kegg.com/dbget-bin/www_bfind_sub?mode=bfind&mode=bfind&max_hit=1000&serv=kegg&dbkey=enzyme&keywords

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EC 4.1.3.1 - isocitrate lyase Mycobacterium tuberculosis glyoxy... DBGET Search Result: ENZYME is...

KEGG Search ENZYME for isocitrate lyase Go Clear

Database: ENZYME - Search term: isocitrate lyase (Total 4 hits)

4.1.3.1
isocitrate lyase; isocitrase; isocitritase; isocitratase; threo-Ds-isocitrate glyoxylate-lyase; isocitrate glyoxylate-lyase

4.1.3.30
methylisocitrate lyase; 2-methylisocitrate lyase; MICL; (2S,3R)-3-hydroxybutane-1,2,3-tricarboxylate pyruvate-lyase

4.2.1.3
aconitate hydratase; cis-aconitase; aconitase; AcnB; 2-methylaconitate hydratase; citrate(isocitrate) hydro-lyase

4.2.1.99
2-methylisocitrate dehydratase; (2S,3R)-3-hydroxybutane-1,2,3-tricarboxylate hydro-lyase

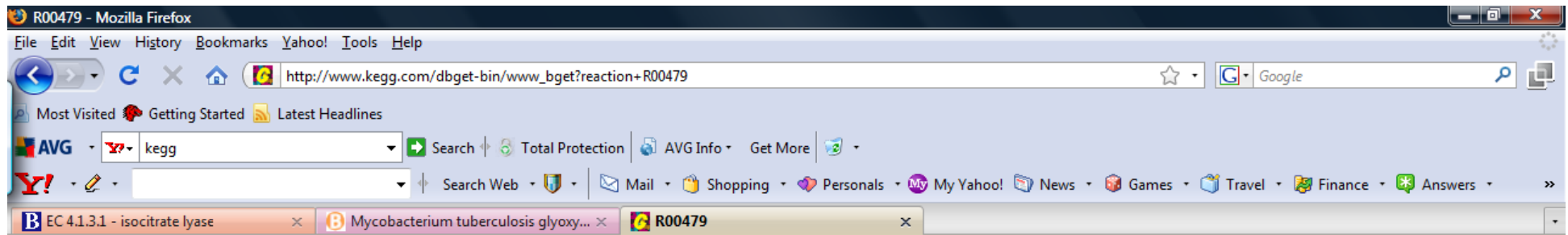
DBGET integrated database retrieval system

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DBGET Search R... 3 Windows Ex... 4 Microsoft Of... 7 Microsoft Of... Cell Biology - A ... Windows Media ... Desktop 12:14 PM

Data Types and Resources

www.kegg.com



KEGG REACTION: R00479 Help	
Entry	R00479 Reaction
Name	isocitrate glyoxylate-lyase (succinate-forming)
Definition	Isocitrate $\rightleftharpoons$ Succinate + Glyoxylate
Equation	C00311 $\rightleftharpoons$ C00042 + C00048
RPair	RP: RP00930 C00042_C00311 main RP: RP00931 C00048_C00311 main
Pathway	PATH: rn00630 Glyoxylate and dicarboxylate metabolism
Enzyme	4.1.3.1
Orthology	KO: K01637 isocitrate lyase [EC:4.1.3.1]
LinkDB	All DBs

DBGET integrated database retrieval system



www.kegg.com



Kinetic Data

- Complexity of equations for describing biochemical reactions makes analytical solutions difficult or infeasible
- Rely on numerical simulations to predict system changes over time and determine steady-state behavior
- Requires numerical data to assign values to system components:
 - K_m , V_{max}
 - Biological half-lives, binding constants
 - Molecule concentrations, diffusion rates
- Some data can be found in specialized DB that are manually curated.
- Many times model developer must curate data for their specific system

Kinetic DBs

- BRENDA (<http://www.brenda-enzymes.info>)
 - Aims to provide comprehensive enzyme information
 - Manually curated DB
 - Large amount of data (Nov. 2007 – 4762 different EC#s, 80,000 different Km values)
 - Multiple search options
 - Molecular weight of enzyme, specific optimal temp, etc.
 - Results downloadable into tab-separated files,
 - Additional tools (EC tree browser, TaxTree)
 - Connected to other DBs (Taxon browsers, PDB, sequence DBs)
- SABIO-RX

Model DBs for Kinetic Parameters

<http://www.brenda-enzymes.org/>

BREND home
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 history
 All enzymes

SEARCH-Navigator
☐ close all ☐ open all
☐ Nomenclature
☐ Reaction & Specificity
☐ Functional Parameters
☐ Isolation & Preparation
☐ Organism related Information
☐ Stability
☐ Enzyme Structure
☐ Disease & References
☐ Application & Engineering

Quick search
 Fulltext search
 Advanced search
 Substructure search
 TaxTree Explorer
 EC Explorer
 Sequence Search
 Genome Explorer
 Ontology Explorer
 Download

BREND
 The Comprehensive Enzyme Information System

TU Braunschweig
 Dept. of Bioinformatics

EC-Number
Enzyme Name
Organism
Protein
Full text
Advanced Search

Display 10 entries

New BREND release will be online soon (July 2009)

Latest paper on BREND (*Nucleic Acids Res.* 37, D588-D592, Jan. 2009):
 BREND, AMEND and FRENDA the enzyme information system: new content and tools in 2009.

Nomenclature	Reaction & Specificity	Functional Parameters
Enzyme Names EC Number Common/ Recommended Name Systematic Name Synonyms CAS Registry Number	Pathway Catalysed Reaction Reaction Type Natural Substrates and Products Substrates and Products Substrates Natural Substrate Products Natural Product Inhibitors Cofactors Metals/Ions Activating Compounds Ligands	Km Value Ki Value IC50 Value pI Value Turnover Number Specific Activity pH Optimum pH Range Temperature Optimum Temperature Range
Isolation & Preparation Purification Cloned Renatured Crystallization		Organism-related information Organism Source Tissue

BRENDA Database

Enzyme Database - BRENDA - Mozilla Firefox

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http://www.brenda-enzymes.org/

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BRENDA home login history All enzymes

SEARCH-Navigator

- close all open all
- Nomenclature
- Reaction & Specificity
- Functional Parameters
- Isolation & Preparation
- Organism related Information
- Stability
- Enzyme Structure
- Disease & References
- Application & Engineering

Quick search Fulltext search Advanced search Substructure search TaxTree Explorer EC Explorer Sequence Search Genome Explorer Ontology Explorer Download

BRENDA
The Comprehensive Enzyme Information System

TU Braunschweig
Dept. of Bioinformatics

[Any question? -> Use the BRENDA Discussion groups](#)

Search Enzyme Names (Synonyms)

isocitrate lyase contains Search show 10 results clear

☒ Recommended Name: exact

EC Number: exact

= amino acid sequences = comprehensive online version = show the catalyzed reaction

Results 1 - 3 of 3

EC Number	Recommended Name	Synonyms	Commentary
4.1.3.1	isocitrate lyase	isocitrate lyase	-
4.1.3.30	Methylisocitrate lyase	2-Methylisocitrate lyase	-
4.1.3.30	Methylisocitrate lyase	Methylisocitrate lyase	-

Results 1 - 3 of 3

download this result as tab stop separated values (Excel, OpenOffice) format

download this result as tab stop separated values (Excel, OpenOffice) format

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Enzyme Dat... Docs Kevin DVD RW Dri... 4 Microsof... 7 Microsof... Cell Biology ... Desktop 6:21 AM

EC 4.1.3.1 - isocitrate lyase - Mozilla Firefox

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http://www.brenda-enzymes.org/php/result_flat.php4?ecno=4.1.3.1&Suchword=&organism[]=Mycobacterium+tuberculosis&show_tr

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[Systematic Name](#)
[Synonyms](#)
[CAS Registry Number](#)
[Reaction](#)
[Reaction Type](#)
Enzyme-Ligand Interactions
[Substrate/Product](#)
[Natural Substrates](#)
[Cofactor](#)
[Metals and Ions](#)
[Inhibitors](#)
[Activating Compound](#)
Functional Parameters
[KM Value \[mM\]](#)
[Ki Value \[mM\]](#)
[IC50 Value \[mM\]](#)
[pI Value](#)
[Turnover Number](#)
[Specific Activity](#)
[pH Optimum](#)
[pH Range](#)
[Temperature Optimum](#)
[Temperature Range](#)
Organism related Information
[Source Tissue](#)

BRENDA
The Comprehensive Enzyme Information System
EC 4.1.3.1 - isocitrate lyase

Information on EC 4.1.3.1 - isocitrate lyase:

This record set is specific for: 1. *Mycobacterium tuberculosis*

Mark a special word or phrase in this record:

Select one or more organisms in this record:

- Kluyveromyces lactis
- Linum usitatissimum
- Lupinus sp.
- Magnaporthe grisea
- Mycobacterium tuberculosis**

Show additional data

- ☒ Do not include text mining results
- ☐ Include **AMENDA** (text mining) results ^{new!} ([more...](#))
- ☐ Include **FRENDA** results ^{new!} (AMENDA + additional results, but less precise; [more...](#))

EC NUMBER	COMMENTARY
4.1.3.1	-

Pathway	KEGG Link
Glyoxylate and dicarboxylate metabolism	00630

Done

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EC 4.1.3.1 - isocitrate lyase - Mozilla Firefox

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http://www.brenda-enzymes.org/php/result_flat.php4?ecno=4.1.3.1&Suchword=&organism[]=Mycobacterium+tuberculosis&show_tr

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Enzyme Nomenclature

- EC number
- Recommended Name
- Systematic Name
- Synonyms
- CAS Registry Number
- Reaction
- Reaction Type
- Enzyme-Ligand Interactions
- Substrate/Product
- Natural Substrates
- Cofactor
- Metals and Ions
- Inhibitors
- Activating Compound

Functional Parameters

- KM Value [mM]
- Ki Value [mM]
- IC50 Value [mM]
- pI Value
- Turnover Number
- Specific Activity
- pH Optimum
- pH Range
- Temperature Optimum
- Temperature Range
- Organism related Information
- Source Tissue

BRENDA
The Comprehensive Enzyme Information System
EC 4.1.3.1 - isocitrate lyase

KM VALUE [mM]	KM VALUE [mM] Maximum	SUBSTRATE	ORGANISM	COMMENTARY	LITERATURE	IMAGE
0.718	-	(2R,3S:2S,3R)-2-methyl-isocitrate	Mycobacterium tuberculosis	ICL1, in 50 mM MOPS pH 6.8, 5 mM MgCl2	681995	-
0.159	-	Isocitrate	Mycobacterium tuberculosis	in 50 mM potassium phosphate, 4 mM MgCl2, 4 mM phenylhydrazine, 12 mM cysteine, pH 7.0, at 37°C	679472	● 2D-image
0.188	-	Isocitrate	Mycobacterium tuberculosis	ICL1, in 50 mM MOPS pH 6.8, 5 mM MgCl2	681995	● 2D-image
1.14	-	Isocitrate	Mycobacterium tuberculosis	ICL2, in Tricine-HCl pH 7.5, 5 mM MgCl2	681995	● 2D-image
0.145	-	threo D-(s)-isocitrate	Mycobacterium tuberculosis	Icl enzyme, 37°C	651648	● 2D-image
1.3	-	threo D-(s)-isocitrate	Mycobacterium tuberculosis	AceA enzyme, Tricine buffer, 37°C, pH 7.5	651648	● 2D-image

Ki VALUE [mM]	Ki VALUE [mM] Maximum	INHIBITOR	ORGANISM	COMMENTARY	LITERATURE	IMAGE
0.003	-	3-nitropropionate	Mycobacterium tuberculosis	Icl enzyme, 37°C	651648	● 2D-image
0.11	-	3-nitropropionate	Mycobacterium tuberculosis	AceA enzyme, 37°C	651648	● 2D-image
0.12	-	Bromopyruvate	Mycobacterium tuberculosis	Icl enzyme, 37°C	651648	● 2D-image
0.71	-	Bromopyruvate	Mycobacterium tuberculosis	AceA enzyme, 37°C	651648	● 2D-image
0.12	-	Itaconate	Mycobacterium tuberculosis	Icl enzyme, 37°C	651648	● 2D-image
0.22	-	Itaconate	Mycobacterium tuberculosis	AceA enzyme, 37°C	651648	● 2D-image
0.19	-	itaconic anhydride	Mycobacterium tuberculosis	Icl enzyme, 37°C	651648	● 2D-image
0.48	-	itaconic anhydride	Mycobacterium tuberculosis	AceA enzyme, 37°C	651648	● 2D-image

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EC 4.1.3.1 - isocitrate lyase - Mozilla Firefox

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Enzyme Nomenclature

- EC number
- Recommended Name
- Systematic Name
- Synonyms
- CAS Registry Number
- Reaction
- Reaction Type
- Enzyme-Ligand Interactions
- Substrate/Product
- Natural Substrates
- Cofactor
- Metals and Ions
- Inhibitors
- Activating Compound
- Functional Parameters
- KM Value [mM]
- Ki Value [mM]
- IC50 Value [mM]
- pI Value
- Turnover Number
- Specific Activity
- pH Optimum
- pH Range
- Temperature Optimum
- Temperature Range
- Organism related Information
- Source Tissue

IC50 VALUE [mM]	IC50 VALUE [mM] Maximum	INHIBITOR	ORGANISM	COMMENTARY	LITERATURE	IMAGE
0.0033	-	itaconic anhydride	Mycobacterium tuberculosis	in 50 mM potassium phosphate, 4 mM MgCl2, 4 mM phenylhydrazine, 12 mM cysteine, pH 7.0, at 37°C	679472	2D-image
0.0477	-	XHD-1	Mycobacterium tuberculosis	in 50 mM potassium phosphate, 4 mM MgCl2, 4 mM phenylhydrazine, 12 mM cysteine, pH 7.0, at 37°C	679472	-
0.0182	-	XHD-2	Mycobacterium tuberculosis	in 50 mM potassium phosphate, 4 mM MgCl2, 4 mM phenylhydrazine, 12 mM cysteine, pH 7.0, at 37°C	679472	-

pI VALUE	pI VALUE MAXIMUM	ORGANISM	COMMENTARY	LITERATURE
5	-	Mycobacterium tuberculosis	calculated from amino acid sequence	651648

TURNOVER NUMBER[1/s]	TURNOVER NUMBER MAXIMUM[1/s]	SUBSTRATE	ORGANISM	COMMENTARY	LITERATURE	IMAGE
5.24	-	Isocitrate	Mycobacterium tuberculosis	ICL1, in 50 mM MOPS pH 6.8, 5 mM MgCl2	681995	2D-image
1.38	-	Isocitrate	Mycobacterium tuberculosis	ICL2, in Tricine-HCl pH 7.5, 5 mM MgCl2	681995	2D-image
1.25	-	(2R,3S:2S,3R)-2-methyl-isocitrate	Mycobacterium tuberculosis	ICL1, in 50 mM MOPS pH 6.8, 5 mM MgCl2	681995	-

SPECIFIC ACTIVITY [μmol/min/mg]	SPECIFIC ACTIVITY MAXIMUM	ORGANISM	COMMENTARY	LITERATURE
1.3	-	Mycobacterium tuberculosis	Icl enzyme	651648

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Biomodel DBs

- BioModels Database
(<http://www.ebi.ac.uk/biomodels-main/>)
 - International effort
 - Provide standards for model curation
 - Define vocabularies for model annotation and connection to biological data resources
 - Provide publicly accessible DB of annotated and curated models in various formats
 - Curated and non-curated models
 - Can browse and run online simulations
 - Can export to various formats (e.g. SBML)

Biomodels Database

www.ebi.ac.uk/biomodels-main

The screenshot shows the BioModels Database website in a Mozilla Firefox browser window. The address bar displays <http://www.ebi.ac.uk/biomodels-main/>. The browser's search bar contains the text "biomodels". The website's header includes the EMBL-EBI logo, a search bar with the text "Enter Text Here", and a "Go" button. Below the header, there are navigation links for "Databases", "Tools", "EBI Groups", "Training", "Industry", "About Us", and "Help". The main content area is titled "BioModels Database - A Database of Annotated Published Models". It features a search bar with buttons for "Search", "Go to the model", and "Advanced search". Below the search bar, there are links for "Browse models", "Simulate in JWS Online", and "Submit a model". The "Browse models" section lists "Curated models (216)", "Browse models using GO", and "Non-curated models (196)". The "Simulate in JWS Online" section has a link to "Simulate in JWS Online". The "Submit a model" section has a link to "Submit a model". The "Model of the month" section for May 2009 features a diagram of a metabolic pathway and text about Folate (Vitamin B7). The "News" section lists two releases: "16th June 2009 Fourteenth release" and "25th March 2009 Thirteenth release". The footer includes a status bar with the date and time, and a taskbar with various application icons.

BioModels Database - A Database of Annotated Published Models

BioModels Database is a data resource that allows biologists to store, search and retrieve published mathematical models of biological interests. Models present in BioModels Database are annotated and linked to relevant data resources, such as publications, databases of compounds and pathways, controlled vocabularies, etc.

Search Go to the model Advanced search

Browse models

- Curated models (216)
- Browse models using GO
- Non-curated models (196)

Simulate in JWS Online

Submit a model

Mirror at California Institute of Technology <http://biomodels.caltech.edu>

BioModels AT SourceForge <http://sourceforge.net/projects/biomodels/>

Model of the month

May, 2009

Folate, also known as Vitamin B7, and its derivatives are important coenzymes in the metabolism of all organisms. Its metabolism is of great interest, as vitamin B7 deficiency can lead to anaemia and may result in birth defects. Furthermore, it has been implicated in cancer and cardiovascular disease.

[Read more...](#)

News

16th June 2009 **Fourteenth release**
[Download All Models Under SBML Format](#)

25th March 2009 **Thirteenth release**
[Download All Models Under SBML Format](#)

Biomodels Database

www.ebi.ac.uk/biomodels-main

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http://www.ebi.ac.uk/biomodels-main/search-models.do?cmd=TEXT:SEARCH

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Search - Models

The search totally returned 5 models.

[New Search](#)

3 Curated Models returned.

BioModels ID	Name	Publication ID	Last Modified
BIOMD0000000093	Yamada2003_JAK_STAT_pathway	12527385	2008-08-21T12:12:24+00:00
BIOMD0000000094	Yamada2003_JAK_STAT_SOCS1_knockout	12527385	2008-08-21T12:12:56+00:00
BIOMD0000000215	Schulz2009_Th1_differentiation	19409816	2009-06-23T16:11:07+00:00

2 Non-curated Models returned.

BioModels ID	Name	Publication ID	Last Modified
MODEL2463683119	Oda2006_TollLikeR	16738560	2007-09-04T16:09:19+00:00
MODEL6399676120	Duarte2007_Homo_sapiens_Metabol_Recon_1	17267599	2008-12-03T11:29:23+00:00

[New Search](#)

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2 Window... DBSolve7.47 2 Microsof... WiggintonKi... MarinoKirc... jacobs2007_r... BioModels D... Desktop 11:12 PM

Biomodels Database

www.ebi.ac.uk/biomodels-main

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BIOMD0000000093 - Yamada2003_JAK_STAT_pathway

SBML formats | Other formats | Actions | [Submit Model Comment/Bug](#)

Model | CellML | SciLab | Math | Physical entities | Parameters | Curation

XPP | BioPAX | VCell | PDF

Reference Publication

Publication ID: [12527385](#)

Cell Res Lett 2003 Jan;534(1-3):190-6.
Control mechanism of JAK/STAT signal transduction pathway.
Yamada S, Shiono S, Joo A, Yoshimura A.
Advanced Technology R&D Center, Mitsubishi Electric Corporation, 8-1-1, Tsukaguchi-Honmachi, Amagasaki, Hyogo 661-8661, Japan.
yamada.satoshi@wrc.melco.co.jp [more](#)

Model

Original Model: **DOQCS**

Submitter: [Sharat Vaytaden](#)

Submission Date: 2007-03-13T22:10:27+00:00

Last Modification Date: 2008-08-21T12:12:24+00:00

Creation Date: 2007-03-07T18:40:22+00:00

Encoders: [Harish Dharuri](#)

set #1 bqbiol:is [Taxonomy Mus musculus](#)
[KEGG Pathway mmu04630](#)
[Gene Ontology JAK-STAT cascade](#)

Notes

NCBS Curation Comments This model shows the control mechanism of Jak-Stat pathway, here SOCS1 (Suppressor of cytokine signaling-1) was identified as the negative regulator of Jak and STAT signal transduction pathway. Note: There are a few ambiguities in the paper like initial concentration of IFN and some reactions were missing in the paper that were employed for obtaining the results. The graphs are almost similar to the graphs as shown in the paper but still some ambiguities regarding the concentration are there. Thanks to Dr Satoshi Yamada for clarifying some of those ambiguities and providing the values

Done

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☐ v37 ☐ SOCS1 unbinding ☐ SOCS1 unbinding ☐ IFNRJ2 dephosphorylation

☐ SOCS1 unbinding ☐ SOCS1 binding ☐ SOCS1 binding ☐ SOCS1 binding

☐ Interferon-receptor binding ☐ IFNR-JAK binding

Physical entities

☐ Compartments	☐ Species		
☐ cytoplasm	☐ Receptor	☐ JAK	☐ Receptor JAK complex
	☐ Interferon-Receptor-JAK complex	☐ IFNRJ dimer	☐ Activated IFNRJ complex
	☐ STAT1c	☐ IFNRJ2_star STAT1c	☐ STAT1c_star
	☐ IFNRJ2_star STAT1c_star	☐ STAT1c_star STAT1c_star	☐ SHP2
	☐ IFNRJ2_star SHP2	☐ PPX	☐ STAT1c_star PPX
	☐ STAT1c STAT1c_star	☐ mRNAC	☐ SOCS1
	☐ IFNRJ2_star SOCS1	☐ IFNRJ2_star SHP2 SOCS1 STAT1c	☐ STAT1c_star STAT1c_star PPX
	☐ IFNRJ2_star SOCS1 STAT1c	☐ IFN	☐ IFNRJ2_star SHP2 STAT1c
	☐ IFNRJ2_star SHP2 SOCS1	☐ IFNR	
☐ nucleus	☐ STAT1n_star STAT1n_star	☐ STAT1n_star	☐ PPN
	☐ STAT1n_star PPN	☐ STAT1n	☐ STAT1n STAT1n_star
	☐ mRNAn	☐ STAT1n_star STAT1n_star PPN	

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SBML formats Other formats Actions Submit Model Comment/Bug

Model Overview Math Physical entities Parameters

Reactions (46)

Receptor-JAK Binding $[JAK] + [Receptor] \rightarrow [Receptor\ JAK\ complex];$

Interferon-Receptor Binding $[Receptor\ JAK\ complex] + [IFN] \rightarrow [Interferon-Receptor-JAK\ complex];$

IFN-Receptor complex dimerization $2.0 \times [Interferon-Receptor-JAK\ complex] \rightarrow [IFNRJ\ dimer];$

IFN-Receptor complex activation $[IFNRJ\ dimer] \rightarrow [Activated\ IFNRJ\ complex];$

Math: $cytoplasm \times kf \times IFNRJ2$ (Detail)

Annotations: set #1 bqboli:is Enzyme Nomenclature 2.7.10.2 KEGG Reaction R02584 Gene Ontology Janus kinase activity

Activated IFNRJ2-STAT1c binding $[Activated\ IFNRJ\ complex] + [STAT1c] \rightarrow [IFNRJ2_star_STAT1c];$

STAT1c activation $[IFNRJ2_star_STAT1c] \rightarrow [STAT1c_star] + [Activated\ IFNRJ\ complex];$

Activated IFNRJ2-STAT1c binding $[STAT1c_star] + [Activated\ IFNRJ\ complex] \rightarrow [IFNRJ2_star_STAT1c_star];$

Activated STAT1c dimerization $2.0 \times [STAT1c_star] \rightarrow [STAT1c_star_STAT1c_star];$

SUP2 binding $[SUP2] + [Activated\ IFNRJ\ complex] \rightarrow [IFNRJ2_star_SUP2];$

MathML - Mozilla Firefox

http://www.ebi.ac.uk/biomodels-main/mathml.do?uri=publ-m

Reaction: INF-Receptor complex activation

rate law: $cytoplasm \times kf \times IFNRJ2$

Compartment

Name	Size
cytoplasm	1.0

Species

Name	Compartment	Initial Amount	Initial Concentration
IFNRJ dimer	cytoplasm		0.0

Parameters

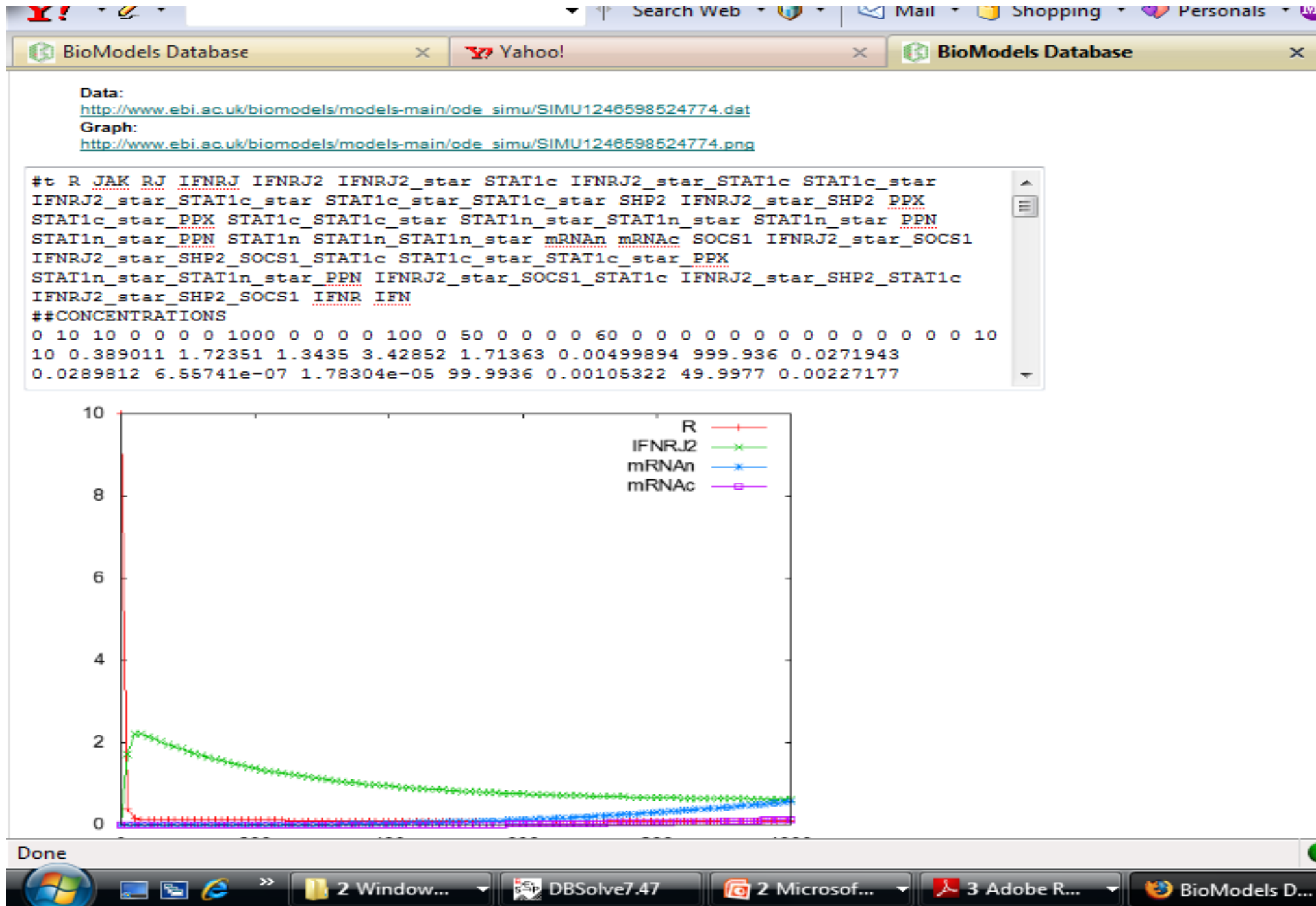
Name	Value
kf	0.0050

Done

Done

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HANDS ON PRACTICE 2:

Model System Using Michaelis-Menten Kinetics

Simulation with OCTAVE*

*Thanks to Daniel Ajuzie for Octave scripts

Dynamic Modeling - II

- Using mass action kinetics:

$$S : \quad \frac{\partial S}{\partial t} \quad = -k_1[E][S] + k_{-1}[ES]$$

$$\begin{aligned} ES : \quad \frac{\partial ES}{\partial t} &= k_1[E][S] - k_{-1}[ES] - k_2[ES] \\ &= k_1[E][S] - (k_{-1} + k_2)[ES] \end{aligned}$$

$$E : \quad \frac{\partial E}{\partial t} \quad = -k_1[E][S] + (k_{-1} + k_2)[ES]$$

$$P : \quad \frac{\partial P}{\partial t} \quad = k_2[ES]$$

Dynamic Modeling - II

Michaelis-Menten Kinetics

- ▶ This gives us:

$$ES = \frac{k_1 E_T S}{k_1 S + (k_{-1} + k_2)} = \frac{E_T S}{S + \frac{(k_{-1} + k_2)}{k_1}}$$

- ▶ Substitute ES:

$$v = \frac{\partial P}{\partial t} = k_2 ES = \frac{k_2 E_T S}{S + \frac{k_{-1} + k_2}{k_1}}$$

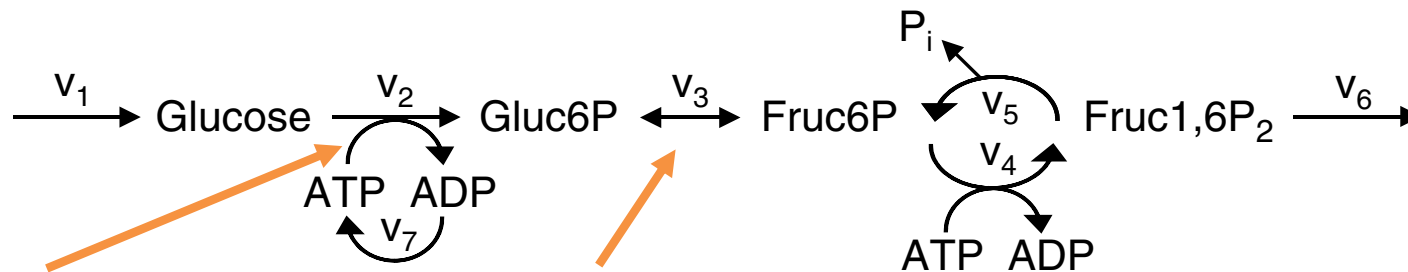
- ▶ Or

$$v = \frac{V_{max} S}{S + K_M}$$


- ▶ Where $V_{max} = k_2 E_T$. Maximal reaction velocity; enzyme is completely saturated with substrate.
- ▶ And $k_2 = k_{cat} = \text{Turnover number}(s^{-1})$
- ▶ $K_M = \frac{k_{-1} + k_2}{k_1}$ is the Michaelis-Menten Constant, which is the substrate concentration that yields half the maximum reaction velocity.

Simulating Dynamic Systems

- Example: Simplified model of upper glycolysis (Klipp et al., Fig. 3.2, Eq. 3.1) **Steps 1 – 3:**



$$\frac{d}{dt} \text{Glucose} = v_1 - v_2$$

$$\frac{d}{dt} \text{Gluc6P} = v_2 - v_3$$

$$\frac{d}{dt} \text{Fruc6P} = v_3 - v_4 + v_5$$

$$\frac{d}{dt} \text{Fruc1,6P}_2 = v_4 - v_5 - v_6$$

$$\frac{d}{dt} \text{ATP} = -\frac{d}{dt} \text{ADP} = -v_2 - v_4 + v_7.$$

$$v = \frac{V_{\max} S}{S + K_M}$$

EC number for v2, v3?

With mass action kinetics, the rate equations read $v_1 = \text{const.} = k_1$, $v_2 = k_2 \cdot \text{Glucose} \cdot \text{ATP}$, $v_3 = k_3 \cdot \text{Gluc6P} - k_{-3} \cdot \text{Fruc6P}$, $v_4 = k_4 \cdot \text{Fruc6P} \cdot \text{ATP}$, $v_5 = k_5 \cdot \text{Fruc1,6P}_2$, $v_6 = k_6 \cdot \text{Fruc1,6P}_2$, and $v_7 = k_7 \cdot \text{ADP}$. Given the values of the parameters k_i , $i = 1, \dots, 7$

Model Simulation with OCTAVE

Model/Function File: glu3.m

```
function dydt = glu3(t, y, k1, k2, k3, nk3, k4, k5,  
k6, k7)
```

```
% For indexing
```

```
glu = 1;
```

```
g6p = 2;
```

```
f6p = 3;
```

```
f16p2 = 4;
```

```
atp = 5;
```

```
adp = 6;
```

```
% Rate equations
```

```
v1 = k1;
```

```
v2 = k2 * y(glu) * y(atp);
```

```
v3 = k3 * y(g6p) - nk3 * y(f6p);
```

```
v4 = k4 * y(f6p) * y(atp);
```

```
v5 = k5 * y(f16p2);
```

```
v6 = k6 * y(f16p2);
```

```
v7 = k7 * y(adp);
```

```
% ODE equations
```

```
dydt(glu) = v1 - v2;
```

```
dydt(g6p) = v2 - v3;
```

```
dydt(f6p) = v3 - v4 + v5;
```

```
dydt(f16p2) = v4 - v5 - v6;
```

```
dydt(atp) = -v2 - v4 + v7;
```

```
dydt(adp) = v2 + v4 - v7;
```

```
dydt = dydt';
```

```
End
```

Model Simulation with OCTAVE

Execution/Script File: solve_glu.m

```
options = odeset('OutputFcn', [], 'AbsTol', 1e-10, 'RelTol', 1e-3);  
[t, y] = ode45(@(t, y) glu3(t, y, 0.25, 1, 1, 1, 1, 1, 1, 2.5), [0, 60],  
[0, 0, 0, 0, 0.5, 0.5], options);  
  
plot(t, y(:, 1));  
hold on;  
plot(t, y(:, 2), 'r');  
plot(t, y(:, 3), 'g');  
plot(t, y(:, 4), 'm');  
plot(t, y(:, 5), 'k');  
plot(t, y(:, 6), 'x-');  
title('Upper glycolysis model');  
ylabel('concentration');  
xlabel('time');  
legend('Glu', 'G6P', 'F6P', 'F16P2', 'ATP', 'ADP');
```

Simulating Dynamic Systems with MATLAB

???

- Plot

